

KARNATAK LAW SOCIETY'S

GOGTE INSTITUTE OF TECHNOLOGY UDYAMBAG, BELGAVI-590008

(An Autonomous Institution under Visvesvaraya Technological University Belagavi)

(APPROVED BY AICTE, NEWDELHI)



Course Project Report

**Topic: Overcurrent Relay Protection Analysis for Different Fault Conditions
in ETAP**

Submitted by

Name	USN	Marks
Toufiq Nadaf	2GI23EE054	

Guide: Prof. Vani Datar

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CERTIFICATE

This is to certify that the Course Project work titled **“Overcurrent Relay Protection Analysis for Different Fault Conditions in ETAP”** carried out by Toufiq Nadaf bearing USN: **2GI23EE054**, has satisfactorily completed the course activity (Seminar/Project) in course **Power System Analysis (22EE62)** under the guidance of **Prof. Vani. DATAR**. It can be considered as a bonafide work carried out in partial fulfilment for the academic requirement of 6th Semester B.E. prescribed by KLS Gogte Institute of Technology, Belagavi during the academic year

2024- 2025.

The report has been approved as it satisfies the academic requirements in respect of Assignment (Course project) prescribed for the said Degree.

Signature of the Faculty Member

Signature of the HOD

Acknowledgement:

I would like to express my sincere gratitude to everyone who contributed to the completion of this report. Firstly, I extend my deepest appreciation to Prof. Vani Datar for their guidance, support, and invaluable insights throughout the process. Their expertise and encouragement were instrumental in shaping the direction and quality of this report.

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Furthermore, I extend my thanks to KLS GIT for providing the necessary resources and facilities to conduct this research effectively.

1.INTRODUCTION

Electrical power systems are subjected to different types of faults due to insulation failure, lightning, equipment malfunction, and human error. Faults in power systems cause abnormal current flow and voltage disturbances, which may damage electrical equipment and interrupt power supply. Therefore, protective devices such as relays and circuit breakers are used to detect and isolate faulty sections quickly and safely.

In this project, different fault conditions such as Three-Phase Fault, Line-to-Ground (L-G) Fault, Line-to-Line (L-L) Fault, and Double Line-to-Ground (L-L-G) Fault are analyzed using ETAP software. The simulation helps in studying fault current magnitude, voltage variation, relay operation, and circuit breaker response during abnormal conditions. The analysis also helps in understanding the importance of overcurrent relay protection in maintaining system safety, reliability, and stability.

❖ Types of Faults

1.Three-Phase Fault

A three-phase fault occurs when all the three phases of the power system are short-circuited together. It is considered the most severe fault because it produces the maximum fault current in the system. During this fault, the overcurrent relay detects the high current and trips the circuit breaker to isolate the faulty section.

Observation

- Maximum fault current is produced.
- System voltage drops suddenly.
- Relay operates quickly.
- Circuit breaker isolates the faulted section.

2. Line-to-Ground (L-G) Fault

A line-to-ground fault occurs when one phase conductor comes in contact with the ground. This is the most common type of fault in power systems. The fault current flows from the phase conductor to the ground through the fault path.

Observation

- Fault current is lower than three-phase fault current.
- Voltage imbalance occurs in the system.
- Overcurrent relay detects abnormal current.
- Faulty phase gets isolated by the circuit breaker.

3. Line-to-Line (L-L) Fault

A line-to-line fault occurs when two phase conductors come into contact with each other. This fault generally occurs due to insulation failure or conductor damage. The fault affects two phases and causes high current flow between them.

Observation

- High current flows between two phases.
- Voltage of affected phases decreases.
- Relay senses the abnormal current condition.
- Circuit breaker disconnects the affected section.

4. Double Line-to-Ground (L-L-G) Fault

A double line-to-ground fault occurs when two phase conductors simultaneously come into contact with the ground. This fault produces high fault current and severe voltage disturbance in the system.

Observation

- Fault current is higher than L-G and L-L faults.
- Two phases become grounded.
- Relay operates to protect the system.
- Circuit breaker isolates the faulty section rapidly.

1.1 Single Line Diagram:

- The Below single line diagram represents a simple electrical power system modeled in ETAP for fault analysis and overcurrent relay protection studies. The diagram consists of a generator, transformer, buses, relay, circuit breaker, and load connected in a single network. The system is designed to study the behavior of the power network during fault conditions and to analyze the operation of protective devices.
- In the system, a 50 MW generator acts as the main power source supplying electrical energy to the network. The generator is connected to Bus1, where the system voltage is approximately 6.95 kV under operating conditions. The generator provides the required power for the connected load and maintains continuous energy flow within the electrical system..
- A 10 MVA transformer (T1) is connected between Bus1 and Bus2. The transformer is used to transfer electrical power between different voltage levels and also provides electrical isolation between sections of the system. The transformer plays an important role in maintaining proper voltage levels and ensuring efficient power transmission throughout the network.
- An overcurrent relay (Relay1) and circuit breaker (CB1) are connected near Bus2 for system protection. The overcurrent relay continuously monitors the current flowing through the system. Under normal operating conditions, the relay remains inactive. When abnormal current due to fault conditions exceeds the preset relay setting, the relay sends a trip signal to the circuit breaker.
- The fault shown in the diagram is a Line-to-Line (L-L) fault occurring near Bus2. During the fault, the system voltage drops to approximately 3.99 kV and the fault current increases to around 13.789 kA. The sequence-of-operation report indicates that Relay1 detects the fault current and operates within a short time interval. After receiving the relay signal, CB1 trips and isolates the faulty section from the healthy part of the system..
- The single line diagram helps in understanding the flow of electrical power, location of faults, relay operation, and circuit breaker action during abnormal conditions. The ETAP simulation provides accurate analysis of fault current magnitude, relay response time, and system performance under fault conditions. This study demonstrates the importance of overcurrent relay protection in maintaining the safety, reliability, and stability of electrical power systems..
- It improves coordination between different protection devices in the electrical network.

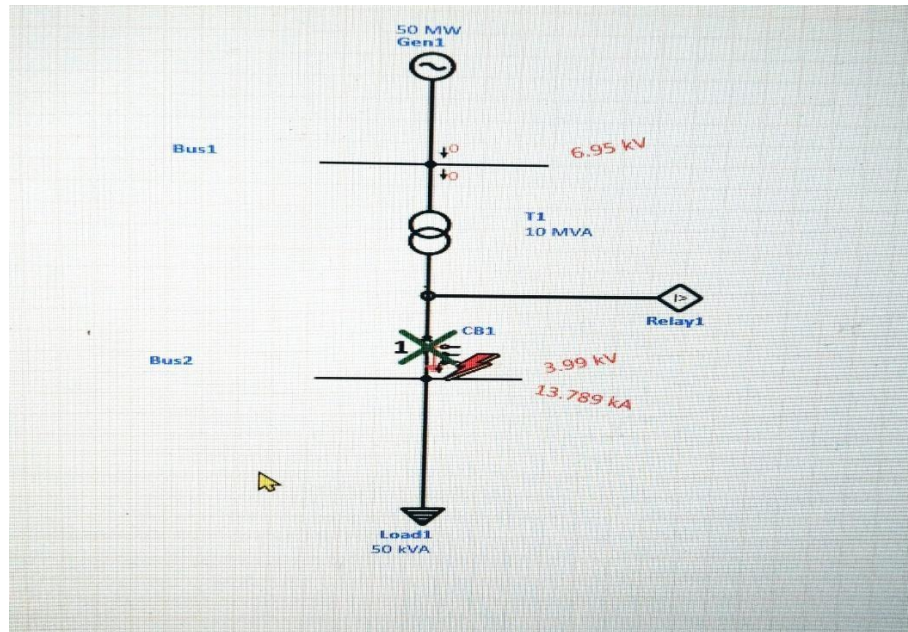


Figure 1: overcurrent relay protection analysis for a three phase fault using etap.

1.2 Objectives:

- The main objective of this project is to design and simulate a power system network using ETAP for analyzing different fault conditions such as Three-Phase Fault, Line-to-Ground Fault, Line-to-Line Fault, and Double Line-to-Ground Fault..
- Another objective is to study the behavior of the electrical power system during abnormal conditions by observing fault current, voltage variation, and the effect of faults on system components such as generators, transformers, buses, and loads.
- The project also aims to analyze the operation and performance of the overcurrent relay and circuit breaker during fault conditions. The relay detects excessive current and sends a trip signal to isolate the faulty section quickly and safely..
- The final objective is to understand the importance of protective relays and proper fault analysis in improving the safety, stability, reliability, and continuity of electrical power system operation..

1.3 Methodology:

- i. The methodology of this project begins with designing the electrical power system in ETAP using a single line diagram. The system consists of a 50 MW generator, buses, a 10 MVA transformer, overcurrent relay, circuit breaker, and load connected properly to form the power network for fault analysis..
- ii. After creating the single line diagram, the ratings and parameters of all system components such as generator voltage, transformer capacity, relay settings, and load values are entered into the ETAP software. The system is first operated under normal conditions to verify proper voltage levels, current flow, and stable operation..
- iii. A Different types of faults such as Three-Phase Fault, Line-to-Ground Fault, Line-to-Line Fault, and Double Line-to-Ground Fault are then applied individually near Bus2 in the system. During fault conditions, the voltage drop, fault current magnitude, and system disturbances are observed carefully through simulation results.
- iv. The overcurrent relay continuously monitors the current flowing through the system and compares it with the preset pickup value. When the fault current exceeds the relay setting, the relay operates and sends a tripping signal to the circuit breaker. The circuit breaker then isolates the faulty section from the healthy part of the system to prevent equipment damage..
- v. The sequence-of-operation report generated in ETAP is analyzed to study relay operating time, breaker tripping time, and overall protection system performance. The obtained results help in evaluating the effectiveness of overcurrent relay protection in detecting and clearing faults quickly and efficiently.
- vi. During the simulation process, the behavior of the system under each fault condition is compared to understand the severity of different faults. The three-phase fault produces the maximum fault current, while Line-to-Ground and Line-to-Line faults create comparatively lower fault currents and voltage disturbances. These comparisons help in identifying the impact of each fault on system performance and protection requirements.
- vii. The complete fault analysis and relay coordination study carried out in ETAP helps in improving the reliability and safety of the electrical power system. The simulation results provide a better understanding of fault detection, relay response, and breaker operation, which are essential for designing an efficient protection system in practical power networks..

2. Observations:

Sl. No.	Fault Type	Normal Condition	Fault Condition	Fault Current
1.	Three-Phase Fault	System operates normally	All three phases short-circuited	13.789 kA
2.	Line-to-Ground (L-G) Fault	Balanced voltage and current	One phase connected to ground	8.25 kA
3.	Line-to-Line (L-L) Fault	Normal power flow	Two phases short-circuited	10.84 kA
4.	Double Line-to-Ground (L-L-G) Fault	Stable system condition	Two phases connected to ground	11.95 kA

3.Recommendations:

1. Proper relay settings should be selected to ensure accurate detection and quick isolation of faults in the power system.
2. Regular maintenance and testing of protective relays, transformers, and circuit breakers should be carried out to improve system reliability..
3. Fault analysis studies using ETAP should be performed periodically to identify possible weaknesses in the protection system..
4. Proper coordination between protective devices should be maintained to isolate only the faulty section without affecting the healthy part of the system.
5. Advanced protection schemes and monitoring systems can be implemented to improve the safety and stability of the power network.

4. Conclusion:

The fault analysis of Three-Phase Fault, Line-to-Ground (L-G) Fault, Line-to-Line (L-L) Fault, and Double Line-to-Ground (L-L-G) Fault was successfully carried out using ETAP software. The simulation results showed that different faults produce different levels of fault current and voltage disturbances in the electrical power system. Among all the faults, the three-phase fault produced the highest fault current and was considered the most severe fault condition. The study also demonstrated the effective operation of the overcurrent relay protection system in detecting abnormal current conditions and isolating the faulty section quickly. The project helped in understanding the importance of fault analysis, relay coordination, and proper protection schemes in maintaining the safety, stability, reliability, and continuous operation of electrical power systems.

5. Reference:

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